 Name: _____

Fall 2016 / EC571

Instructor:Ajay Joshi

Exam 2

Date: November 3, 2016

Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
Bonus	5	
Total	100	

1. The duration of the exam is 2 hours.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 6 questions. Please indicate the question number that should not be graded.
5. The solutions should be written in the spaces provided on the test sheet.
6. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
7. Material sharing is not allowed.
8. Calculators are allowed.
9. Boston University Academic Code of Conduct applies.

Question 1

Sketch the schematics for $f(A,B,C,D) = \overline{A \cdot B + C \cdot D}$ using each of the following circuit techniques. You can assume that you have both complemented and non-complemented inputs:

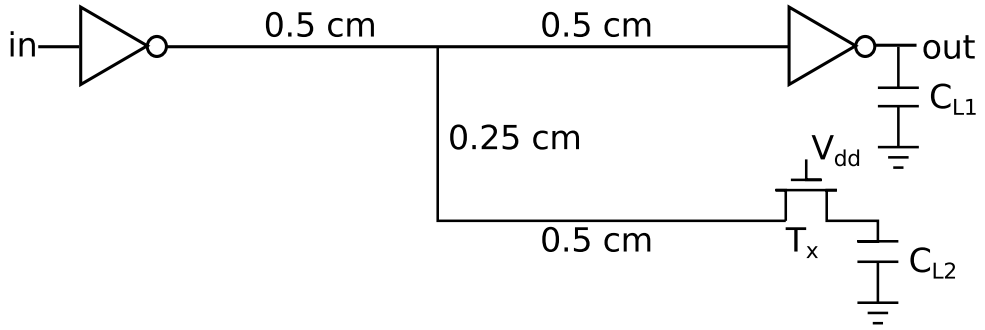
1. Static CMOS. (5 points)

2. Pseudo-nMOS. (5 points)

3. Dynamic. (5 points)

4. Pass transistors (including the inverter at the end). (5 points)

Question 2



1. For the circuit shown in the above figure, draw the equivalent RC network that includes all the parts. Assume a lumped model for each interconnect segment. (6 points)

2. Write the Elmore delay equations for the entire circuit from in to out. (8 points)

3. Calculate the Elmore delay for the entire circuit. You can use the following technology parameters for calculating the delay - V_{dd} = 1.5 V, L = 0.25 micron $R_{sq(NMOS)}$ = 10 Kohm/sq, $R_{sq(PMOS)}$ = 25 Kohm/sq, C_g = 1 fF/micron, C_d = 1.5 fF/micron, wire capacitance = 0.25 fF/micron, wire resistance = 15 ohms/sq and width of transistor T_x = 500 nm. The interconnect has a width of 0.5 micron. Assume that all inverters use a minimum width of 300 nm for the NMOS. You will need to choose the width of the PMOS transistor such that the low-to-high delay of the inverters matches with the high-to-low delay. Assume a C_{L1} = 20 fF and C_{L2} = 10 fF. (8 points)

Question 3

For the circuit shown in Figure 1, sketch the timing waveforms of the voltages at Vout, x, y and z for the given inputs A, B, C (Table 1). Also calculate the energy drawn from the power supply during each half of the clock cycles over the 5 clock cycles listed in Table 1. Here Tclk is the clock period. Please assume that clk, A, B and C are ramp signals and change at the time stamps given in the table. You can approximate the time scale, but be sure to compute the voltage levels. Assume V_{DD} = 2.5 V, V_T = 0.5 V, C_W = 2 fF. C_X = 3 fF, $C_Y = 2.5fF$ and C_Z = 2 fF. Also assume that the clock period is long enough for discharging the all capacitors. (20 points)

Time	Clk	A	B	C	D
0	0	0	0	0	0
Tclk/2	1	1	0	0	1
Tclk	0	0	1	1	0
3Tclk/2	1	1	1	0	1
2Tclk	0	1	0	1	0
5Tclk/2	1	1	1	1	0
3Tclk	0	1	1	1	1
7Tclk/2	1	0	1	0	0
Tclk	0	0	0	1	1
9Tclk/2	1	0	1	1	0
5Tclk	0	1	1	0	1

Table 1

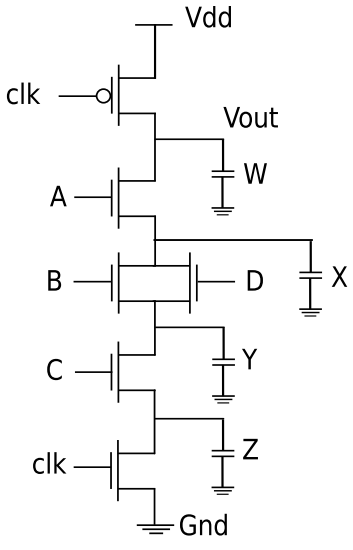


Figure 1

Question 4

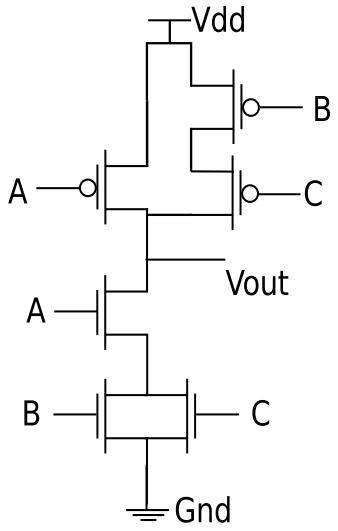


Figure 2

1. Determine sizes of the six transistors in the above figure so that the circuit provides the same pull-up and pull-down current at the output as a unit inverter. The input capacitance of all inputs (A, B, and C) should be the same. The width ratio of PMOS to NMOS in the unit inverter is 2:1. Hint: Start with the pull-up network. (7 points)

2. Determine the logical effort of the resulting gate with respect to input A. (6 points)

3. Construct the truth table for the gate in part 1 and compute the probability of the energy consuming transitions of the output, if the input probabilities are p(A = 0) = 0.2, p(B = 0) = 0.5, p(C = 0) = 0.4. (7 points)

Question 5

1. Explain the working of a negative edge-triggered flip-flop and a positive level-triggered latch. (5 points)

2. Design a negative edge-triggered flip-flop using two latches. (5 points)

3. Explain the concepts of negative setup time and negative hold time (5 points)

4. Explain how clock-to-q delay of a flip-flop increases as we start pushing the input data edge closer to the clock edge. You can use the flip-flop circuit that we discussed in class to answer this question. (5 points)

Question 6

1. Compute the probability of the energy consuming transitions of the output for logic function $f_1 = \overline{A \cdot B + C \cdot D}$ implemented in static complementary CMOS logic and dynamic logic. The probability of having 0 as the input at A, B, C and D is as follows: P(A=0) = 0.2, P(B=0) = 0.3, P(C=0) = 0.8 and P(D=0) = 0.6. *Hint: You need to determining the probability of 0 to 1 transition for both logic style implementations of the function. (10 points)*

2. Design the following functions as multi-output domino logic $f_1 = (P \cdot (Q+R)) \cdot S$, $f_2 = P \cdot (Q+R)$ and $f_3 = Q+R$. (5 points)

3. Design $f = A \cdot B + C$ and its complement using dual rail domino logic. (5 points)

Bonus question (5 points)
Prof. Joshi is thinking of visiting some place warm over the winter break. Please suggest 5 locations around the world where he can travel to enjoy some warm weather in December.